

Domain Electricity

code	service	✓ 1						
E-2	Reporting information regarding local electricity generation	Service group: Feedback - Reporting information						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	None	0	0	0	0	0	0	0
level 1	Current generation data available	1	0	0	0	0	1	1
level 2	Actual values and historical data	1	0	0	0	0	1	2
level 3	Performance evaluation including forecasting and/or benchmarking	1	0	0	0	0	1	3
level 4	Performance evaluation including forecasting and/or benchmarking; also including predictive management and fault detection	1	0	0	1	0	2	3
Information sources Standard?		IEC 61850-7-420	IEC 61850-7-420					
code	service	✓ 1						
E-3	Storage of (locally generated) electricity	Service group: DER - Storage						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	None	0	0	0	0	0	0	0
level 1	On site storage of electricity (e.g. electric battery)	0	1	0	2	0	0	0
level 2	On site storage of energy (e.g. electric battery or thermal storage) with controller based on grid signals	0	2	0	2	0	0	0
level 3	On site storage of energy (e.g. electric battery or thermal storage) with controller optimising the use of locally generated electricity	0	2	0	2	0	0	0
level 4	On site storage of energy (e.g. electric battery or thermal storage) with controller optimising the use of locally generated electricity and possibility to feed back into the grid	0	3	0	2	0	0	0
Information sources Standard?								

E-8	Support of (micro)grid operation modes	Service group: DSM- Storage						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	None	0	0	0	0	0	0	0
level 1	Automated management of (building-level) electricity consumption based on grid signals	0	2	0	2	0	0	0
level 2	Automated management of (building-level) electricity consumption and electricity supply to neighbouring buildings (microgrid) or grid	0	2	0	2	0	0	0
level 3	Automated management of (building-level) electricity consumption and supply, with potential to continue limited off-grid operation (island mode)	0	3	0	3	0	0	0
level 4	0							
Information sources								
Standard?		IEC TC 57	IEC TC 57				IEC TC 57	IEC TC 57

code	service	1						
E-11	Reporting information regarding energy storage	Service group: Feedback - Reporting information						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	None	0	0	0	0	0	0	0
level 1	Current state of charge (SOC) data available	1	0	0	0	0	1	1
level 2	Actual values and historical data	1	0	0	0	0	1	2
level 3	Performance evaluation including forecasting and/or benchmarking	1	0	0	0	0	1	3
level 4	Performance evaluation including forecasting and/or benchmarking; also including predictive management and fault detection	1	0	0	1	0	2	3
Information sources								
Standard?								

code	service	1						
E-12	Reporting information regarding electricity consumption	Service group: Feedback - Reporting information						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	None	0	0	0	0	0	0	0
level 1	reporting on current electricity consumption on building level	0	0	0	0	0	0	1
level 2	real-time feedback or benchmarking on building level	1	0	0	0	0	0	2
level 3	real-time feedback or benchmarking on appliance level	2	0	0	0	0	1	3
level 4	real-time feedback or benchmarking on appliance level with automated personalized recommendations	3	0	0	1	0	2	3
Information sources								

		Standard?							
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